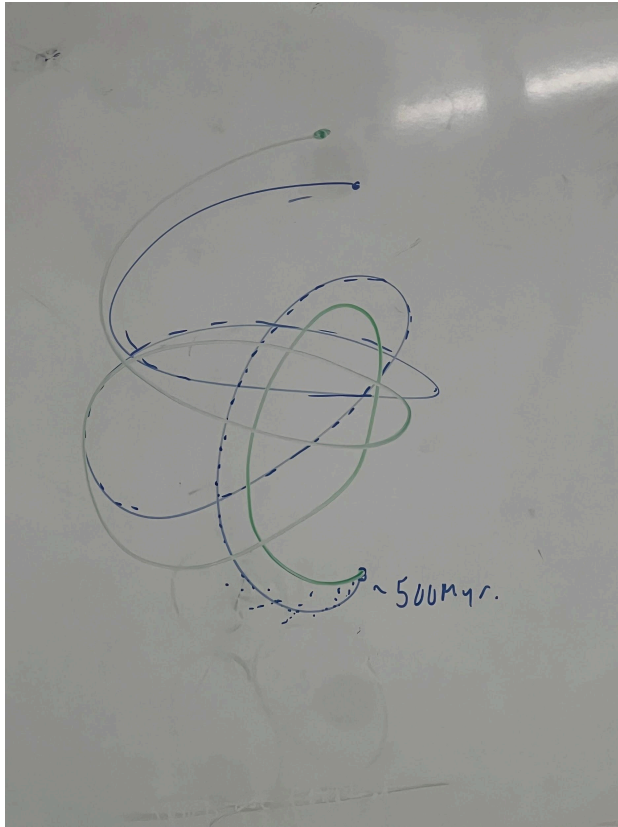


Tasks:

Think of two functions for Joe.

1. Calculate the Kings Model
2. Calculate the planar space (initial position & velocity) from chen paper parameters

Current Problem:**Notes:**

particles
no mass.

$$\oint = \frac{P.E.}{m \omega}$$

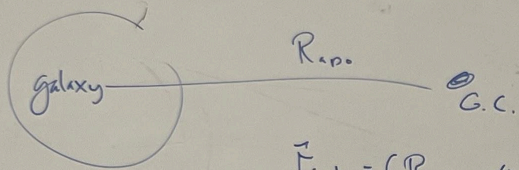
$$P.E. = m \oint$$

$$L = R_a V_a = R_p V_p \Rightarrow V_p = \frac{R_a}{R_p} V_a \Rightarrow V_p^2 = \frac{R_a^2}{R_p^2} V_a^2$$

$$\frac{1}{2} V_p^2 + \oint(R_p) = \frac{1}{2} V_a^2 + \oint(R_a)$$

$$\frac{1}{2} V_a^2 - \frac{1}{2} V_p^2 = \oint(R_p) - \oint(R_a)$$

$$\frac{1}{2} V_a^2 - \frac{1}{2} V_a^2 \frac{R_a^2}{R_p^2} = \oint(R_p) - \oint(R_a)$$



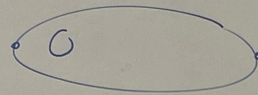
$$\vec{r}_{intn} = (R_{apo}, 0, 0)$$

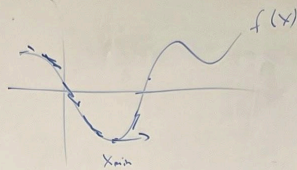
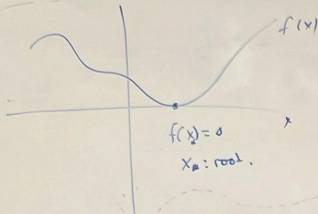
previous used the
particle spray

* hack

× no actual mass to
any of the particles

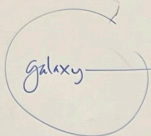
⇒ there is no real
mass loss rate.





$$\frac{df}{dx}(x_{\min}) = 0$$

previous used the particle spray



R_{apo}

G.C.

$$g(x) = \frac{d\phi}{dx}$$

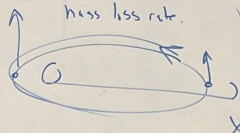
* hack

no actual mass to $\vec{a}_g$ of the particles

$\Rightarrow$ there is no real mass loss rate.

$$\vec{r}_{\text{intin}} = (R_{\text{apo}}, 0, 0)$$

$$\vec{V} = (0, V_{\text{az}}, 0)$$



$$V_{\text{az}} = \frac{1}{2e} [\phi(R_{\text{a}}) - \phi(R_{\text{p}})] (1-e)^{-1}$$

$$V_{\text{a}} = (1-e) \sqrt{\frac{\phi(R_{\text{a}}) - \phi(R_{\text{p}})}{2e}}$$

$$(R_{\text{a}} + R_{\text{p}})$$

$$R_{\text{a}} e$$

$$R_{\text{a}}$$

$$R_{\text{p}}$$

$$e =$$

$$L =$$

$$\text{Spec}$$

$$\frac{1}{2} V$$

$$= \frac{P.E.}{m_{\text{miss}}}$$

$$E_i = m \Phi$$

$$\Rightarrow V_P = \frac{R_a}{R_P} v_a \Rightarrow V_P^2 = \frac{R_a^2}{R_P^2} v_a^2$$

$$V_a^2 \left(1 - \frac{R_a^2}{R_P^2} \right) = 2 [\Phi(R_P) - \Phi(R_a)]$$

$$(R_a)$$

$$V_a^2 \left(1 - \frac{(1+e)^2}{(1-e)^2} \right) = 2 [\Phi(R_P) - \Phi(R_a)]$$

$$R_P) - \Phi(R_a)$$

$$V_a^2 \left(\frac{(1-e)^2 - (1+e)^2}{(1-e)^2} \right) = \dots$$

$$R_P) - \Phi(R_a)$$

$$V_a^2 \left(1 - \frac{1+2e+e^2}{1-2e+e^2} \right)$$

$$= (1-e)^2$$

$$= 2 [\Phi(R_P) - \Phi(R_a)]$$

$$-V_a^2 4e = 2 [\Phi(R_P) - \Phi(R_a)]$$

$$f\left(1 + \frac{\Delta r}{2}\right) =$$

$$\vec{F} = \frac{d\vec{p}}{dt}, \quad \vec{p} : \text{momentum} = \frac{1}{r^2} \pm \left[\dots \right]$$

$$\vec{p} = m\vec{v}$$

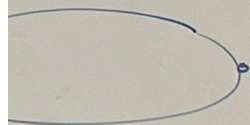
$$= m \frac{d\vec{v}}{dt} + \left(\frac{dm}{dt} \vec{v} \right)$$

$$m \frac{d\vec{v}}{dt} : \vec{F} - \frac{dm}{dt} \vec{v}$$

$$\vec{a} = \frac{\vec{F}}{m} - \underbrace{\frac{\dot{m}}{m} \vec{v}}_{\text{term}}, \quad m \neq \frac{dm}{dt}$$

used the
e spray

and mass to
of the particles
there is no real
mass loss rate.



$$(R_a + R_p)e = R_a - R_p$$

$$R_a e - R_a = -R_p - R_p e$$

$$R_a(e-1) = -R_p(1+e)$$

$$\frac{R_a}{R_p} = \frac{(1+e)}{(1-e)}$$

$$e = \frac{R_a - R_p}{R_a + R_p}$$

All particles
are lost particles
⇒ no mass.

$$L = r v \quad \text{specific angular momentum}$$

Specific Energy

$$\frac{1}{2} v^2 + \Phi(r)$$

$$L = R_a v_a$$

$$\frac{1}{2} v_p^2 + \Phi(r)$$

$$\frac{1}{2} v_a^2$$

$$\frac{1}{2} v_a^2$$